Xinhe (Olivia) Huang

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RESEARCH INTERESTS

Agentic AI and foundation models for biological discovery; AI-guided protein design and genome engineering; multimodal learning across biological scales.

EDUCATION

Huazhong University of Science and Technology — Wuhan, China
Ph.D. in Biomedical Engineering, Advisor: Prof. Yu Xue — *Sep. 2022 – Present*

Massachusetts Eye and Ear, Harvard Medical School — Boston, MA, USA
Visiting Ph.D. Student, Advisor: Prof. Mengyu Wang — *Feb. 2026 – Jul. 2026*

Huazhong University of Science and Technology — Wuhan, China
B.S. in Bioinformatics — *Sep. 2018 – Jun. 2022*

SELECTED PUBLICATIONS

- **Huang, X.**†, Qin, J.†, Tang, F.†, et al. "XunZi, an AI biologist, reveals disease-modifying targets." ***Nature Biomedical Engineering***. 2026; doi:10.1038/s41551-026-01769-6.
- Qin, J.†, **Huang, X.**†, Gou, S.†, et al. "Ketogenic diet reshapes cancer metabolism through lysine β-hydroxybutyrylation." ***Nature Metabolism***. 2024;6(8):1505–1528.
- **Huang, X.**†, Feng, Z.†, Liu, D., et al. "PTMD 2.0: an updated database of disease-associated post-translational modifications." ***Nucleic Acids Research***. 2025;53(D1):D554–D563.
- Peng, J.†, **Huang, X.**†, Yang, K., et al. "dbPTH: A Comprehensive Database for Protein Targets of Herbal Ingredients." ***Genomics, Proteomics & Bioinformatics***. 2026; doi:10.1093/gpbjnl/qzag040.
- Peng, D.†, Zheng, L.†, Liu, D.†, **Huang, X.**, et al. "Large-language models facilitate discovery of molecular signatures regulating sleep and activity." ***Nature Communications***. 2024;15(1):3685.

RESEARCH EXPERIENCE

Ph.D. Researcher — Huazhong University of Science and Technology
Biomedical Engineering and Computational Biology — *Sep. 2022 – Present*

- Developed **agentic AI systems** combining biomedical LLMs, multi-omics foundation models, and genomic agents to generate disease-modifying target hypotheses and annotate regulatory elements.
- Curated a **5M-entry deep mutational scanning atlas** and developed **protein language models** to learn functional fitness landscapes for AI-guided protein design.
- Built AI pipelines for ketogenic-diet cancer metabolism, sleep and activity regulators, PTM-associated disease databases, and ingredient–target interaction resources.
- Led computational and experimental collaborations from AI-generated hypotheses to experimental validation and biological interpretation.

Visiting Ph.D. Researcher — Massachusetts Eye and Ear, Harvard Medical School
Multimodal Genomic Foundation Models — *Feb. 2026 – Jul. 2026*

- Developed contrastive learning methods to align genome representations with biological knowledge for functional annotation and single-cell analysis.

CONFERENCE PRESENTATIONS

Cell Symposium: Metabolites — Organellar Communication in Health and Disease, Sitges, Spain, May 2025 (Poster).

Nature Conference: AI-Augmented Biology, Nanjing, China, Oct. 2025 (Poster).

TECHNICAL SKILLS

AI & ML: LLM continual pretraining, post-training, and parameter-efficient fine-tuning (LoRA/PEFT); RAG and multi-agent workflows; contrastive, multimodal, and state-space sequence modeling

Biological Data Science: large-scale biological data curation and integration; multi-omics and single-cell analysis

Programming & Engineering: Python, R, Bash, PyTorch, JAX, Hugging Face Transformers; distributed training (DDP/FSDP); SLURM/HPC; scientific software, database, and web platform development